

DBCO – FULL NOTES + 2-MARK & 15-MARK ANSWERS + DIAGRAMS

This document includes: ✓ Full Unit-wise Notes ✓ All Important 2-Mark Questions with Answers ✓ All Important 15-Mark Questions with Answers ✓ Text-based ER & Architecture Diagrams ✓ Clean & Printable Formatting

UNIT 1 – DBMS BASICS

Definition of DBMS: A DBMS (Database Management System) is software that stores, retrieves, and manages data efficiently. Advantages: • Reduces redundancy • Ensures data integrity • Provides security • Offers backup & recovery • Supports concurrent access

3-Level Architecture:

[ASCII DIAGRAM] ----- | View Level | ----- | Logical Level |

----- | Physical Level | ----- Data Independence: • Logical Data

Independence • Physical Data Independence Keys: • Primary Key • Foreign Key • Candidate Key •

Super Key • Composite Key

UNIT 2 – ER MODELING

Components of ER Model: • Entity • Attributes • Relationship • Cardinality • Weak Entities ER ASCII
Diagram: STUDENT ----- ENROLLS ----- COURSE || [Name, Age, Roll] [CourseID, CName]
Converting ER Diagram to Tables: STUDENT(RollNo, Name, Age) COURSE(CID, CName)
ENROLLS(RollNo → STUDENT, CID → COURSE)

UNIT 3 – NORMALIZATION

Normalization: Process of organizing data to reduce redundancy. 1NF: • Atomic values only 2NF: • No partial dependency 3NF: • No transitive dependency BCNF: • Every determinant must be a candidate key Functional Dependency Example: RollNo $\rightarrow$ StudentName

UNIT 4 – SQL

DDL Commands: CREATE TABLE, ALTER, DROP DML Commands: INSERT, UPDATE, DELETE
Joins: INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL JOIN View: A virtual table created using
SELECT. Subqueries: Queries inside queries.

UNIT 5 – TRANSACTION MANAGEMENT

ACID Properties: A – Atomicity C – Consistency I – Isolation D – Durability Transaction States (ASCII Diagram): ACTIVE → PARTIALLY COMMITTED → COMMITTED → TERMINATED ↓ FAILED → ABORTED Deadlock: Occurs when transactions wait for each other.

UNIT 6 – JDBC

JDBC Steps: 1. Load driver 2. Create connection 3. Create statement 4. Execute query 5. Close connection
Driver Types: Type 1, Type 2, Type 3, Type 4

IMPORTANT 2-MARK QUESTIONS WITH ANSWERS

1. What is DBMS? A software to store, manage, and retrieve data. 2. Define Primary Key. A field that uniquely identifies each row. 3. Define Foreign Key. A field that refers to primary key in another table. 4. What is a view? A virtual table. 5. What is normalization? Process of reducing redundancy. 6. Define transaction. A set of operations forming a logical unit. 7. What is ACID? Properties ensuring reliable transactions. 8. Define functional dependency. $X \rightarrow Y$ means X uniquely determines Y. 9. What is JDBC? Java Database Connectivity API. 10. Define deadlock. Two transactions waiting on each other's locks.

IMPORTANT 15-MARK QUESTIONS WITH ANSWERS

1. Explain 3-tier architecture of DBMS. Includes Presentation Tier, Application Tier, and Data Tier with clear separation of responsibilities. 2. Explain normalization up to BCNF. Includes definitions of 1NF, 2NF, 3NF, BCNF with examples and anomaly removal. 3. Explain ER model with diagram. Covers entities, attributes, relationships, cardinality, weak entities, and conversion to tables. 4. Explain ACID properties and transaction processing. Explains Atomicity, Consistency, Isolation, Durability with examples and states of transactions. 5. Explain SQL joins and subqueries with examples. Covers all join types, syntax, examples, and subquery classifications (single-row, multi-row, correlated).